

SemEval 2026 Task 5: Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding

1. Problem Description

This problem refers to the task of Word Sense Disambiguation, with a little twist. Usually, it is assumed that only one word sense is the correct one, but in our case, we want to grade the possible senses we are given; that is because in reality, ambiguities, underspecification and subjective opinions can influence which word senses seem more plausible in a given context, and there is a difference between the intuition of humans and the prediction of computational models.

More to the point, the task is to estimate how plausible a given word sense of a target homonym is, using a short story as context. As specified above, rather than trying to ‘find’ a single correct meaning, for each story and each given word, along with its plausible senses, we must predict how plausible each sense is with regards to the context. The plausibility will be measured on a scale from 1 to 5, reflecting how well each interpretation fits with the story.

The dataset is based on human judgements provided by multiple annotators, who rated the plausibility of each word-sense-story pair. Finally, the goal of the task is to model these human plausibility ratings, capturing the preferred sense and how strongly it is supported by the narrative context.

2. Existing Approaches

A. Graded Word Sense Assignment - Erk & McCarthy [1]

Erk and McCarthy studied graded human sense ratings rather than discrete choices. Annotators rated how applicable each sense is on a scale, thus creating their dataset. Using said dataset, they concentrated on training supervised models and compared them against a baseline model that simply used training data to observe a probability of distribution over senses regardless of context. Along with these models, vector space models have been evaluated, using training data, to locate a word sense in semantic space.

The following models are being discussed:

Single-best-sense WSD (WSD/single). As the name suggests, this model assigns to each test occurrence of a target word, a single best-fitting sense. Thus, one word sense gets graded with 1, and the rest with 0.

WSD Confidence level as judgement (WSD/conf). Similar to the model above, but it is used to predict a judgement for each sense of a target occurrence, taking the confidence level returned by each sense-specific binary classifier as the predicted judgment.

Word senses as points in semantic space (Prototype). A model of word senses as prototypes is used, where each is represented as a point in the semantic space. This model measures the distance between feature vectors in space. A single vector for each sense is computed. The predicted judgement for a test sentence and a sense is the similarity of the sentence’s vector to the centroid vector for the sense, computed using cosine.

Results: WSD/single - below average. WSD/conf - slightly above baseline. Prototype - higher precision than WSD/conf

Model	by lemma			by lemma+sense			by lemma+sentence			J/S	P	R	F
	ρ	*	**	ρ	*	**	ρ	*	**				
WSD/single	0.267	87.5	75.0	0.053	6.3	4.2	0.28	2.8	1.8	0.39	58.7	25.5	35.5
WSD/conf	0.396	87.5	87.5	0.177	33.3	18.8	0.401	10.8	3.0	0.164	81.8	37.1	51.0
Prototype	0.245	62.5	62.5	0.053	20.8	8.3	0.396	15.3	2.5	0.173	58.4	78.3	66.9
Prototype/2	0.292	87.5	87.5	0.086	14.6	4.2	0.478	22.8	7.5	0.164	68.2	63.3	65.7
Prototype/N	0.396	100.0	100.0	0.137	22.9	14.6	0.396	15.3	2.5	0.173	82.2	29.9	43.9
Prototype/2N	0.465	100.0	100.0	0.168	29.8	23.4	0.478	22.8	7.5	0.164	82.6	30.9	45.0
baseline	0.338	87.5	87.5	0.0	0.0	0.0	0.355	10.3	3.0	0.167	79.9	34.5	48.2

Table 5: Evaluation: computational models, and baseline. *, **: percentage significant at $p \leq 0.05$, $p \leq 0.01$

B. BERT for Word Sense Disambiguation with Gloss Knowledge [2]

In this paper, the authors propose 3 BERT-based models for word sense disambiguation. The focus is on how to better leverage gloss information in a supervised neural WSD system. Context-gloss pairs are constructed from glosses of all possible senses of the target word (in Word-Net). The WSD task is thus treated as a sentence-pair classification problem.

Context-Gloss Pairs of the target word [research]		Label	Sense Key
[CLS] Your research ... [SEP] systematic investigation to ... [SEP]		Yes	research%1:04:00::
[CLS] Your research ... [SEP] a search for knowledge [SEP]		No	research%1:09:00::
[CLS] Your research ... [SEP] inquire into [SEP]		No	research%2:31:00::
[CLS] Your research ... [SEP] attempt to find out in a ... [SEP]		No	research%2:32:00::
Context-Gloss Pairs with weak supervision of the target word [research]		Label	Sense Key
[CLS] Your “research” ... [SEP] research: systematic investigation to ... [SEP]		Yes	research%1:04:00::
[CLS] Your “research” ... [SEP] research: a search for knowledge [SEP]		No	research%1:09:00::
[CLS] Your “research” ... [SEP] research: inquire into [SEP]		No	research%2:31:00::
[CLS] Your “research” ... [SEP] research: attempt to find out in a ... [SEP]		No	research%2:32:00::

Table 1: The construction methods. The sentence is taken from SemEval-2007 WSD dataset. The ellipsis “...” indicates the remainder of the sentence or the gloss.

The pre-trained BERT model is fine-tuned, and 3 GlossBERT models are resulted:

GlossBERT(Token-CLS) - Context-Gloss pairs are used as input. The model locates the token/tokens of the ambiguous word, and takes the final state of that token. The model

learns the answer to the question “Given this word in this context, and given this gloss, do they match”.

GlossBERT (Sent-CLS) - The same input is fed to the model, but now the model takes the final hidden state of the first token. The classifier learns to answer the question “Does this sentence as a whole match this gloss?”.

GlossBERT (Sent-CLS-WS) - The model is fed the same input, but this time, it is told (indirectly) which word in the sentence is the ambiguous one (weakly supervised). The model thus learns to answer the question “Does the gloss match this sentence with regards to this highlighted word?”.

System	Dev	Test Datasets				Concatenation of Test Datasets				
	SE07	SE2	SE3	SE13	SE15	Noun	Verb	Adj	Adv	All
MFS baseline	54.5	65.6	66.0	63.8	67.1	67.7	49.8	73.1	80.5	65.5
Lesk _{ext+emb}	56.7	63.0	63.7	66.2	64.6	70.0	51.1	51.7	80.6	64.2
Babelify	51.6	67.0	63.5	66.4	70.3	68.9	50.7	73.2	79.8	66.4
IMS	61.3	70.9	69.3	65.3	69.5	70.5	55.8	75.6	82.9	68.9
IMS _{+emb}	62.6	72.2	70.4	65.9	71.5	71.9	56.6	75.9	84.7	70.1
Bi-LSTM	-	71.1	68.4	64.8	68.3	69.5	55.9	76.2	82.4	68.4
Bi-LSTM _{+att.+LEX+POS}	64.8	72.0	69.1	66.9	71.5	71.5	57.5	75.0	83.8	69.9
GAS _{ext} (Linear)	-	72.4	70.1	67.1	72.1	71.9	58.1	76.4	84.7	70.4
GAS _{ext} (Concatenation)	-	72.2	70.5	67.2	72.6	72.2	57.7	76.6	85.0	70.6
CAN ^s	-	72.2	70.2	69.1	72.2	73.5	56.5	76.6	80.3	70.9
HCAN	-	72.8	70.3	68.5	72.8	72.7	58.2	77.4	84.1	71.1
BERT(Token-CLS)	61.1	69.7	69.4	65.8	69.5	72.0	57.8	73.5	84.4	68.6
GlossBERT(Sent-CLS)	69.2	76.5	73.4	75.1	79.5	79.1	65.4	79.3	84.8	75.8
GlossBERT(Token-CLS)	71.9	77.0	75.4	74.6	79.3	78.8	66.8	79.9	85.0	76.3
GlossBERT(Sent-CLS-WS)	72.5	77.7	75.2	76.1	80.4	79.8	67.1	79.6	87.4	77.0

Table 3: F1-score (%) for fine-grained English all-words WSD on the test sets in the framework of Raganato et al. (2017b) (including the development set **SE07**). **Bold** font indicates best systems. The five blocks list the MFS baseline, two knowledge-based systems, two traditional *word expert* supervised systems, six recent neural-based systems and our systems, respectively. Results in first three blocks come from Raganato et al. (2017b), and others from the corresponding papers.

Author Observations:

- BERT Base model performs slightly better than the BERT Large model.
- The total number of parameters for the pre-trained model is 110Million.
- The GlossBERT(Token-CLS) performs better than GlossBERT(Sent-CLS), which proves that highlighting the target word in the sentence is meaningful.
- The GlossBERT(Sent-CLS-WS) performs best in most circumstances, which might be the result of combining the 2 previous methods.

C. Organizer tests on several LLM’s [3]

The organizers of the contest for this task tested their dataset on several LLM’s. The authors found that some models perform pretty well, but still below the human upper bound. The best performing model is DeepSeek-V3, the only model to obtain an accuracy of more than 80%. The second best model o3 performs second best, with an accuracy of 76%, followed by the GPT-4o models.

	Spearman	Acc. w/in SD
Random	0.000	0.454
Majority	N/A	0.558
Llama-3 (0-shot)	0.462	0.663
Mistral (0-shot)	0.382	0.568
Mixtral (0-shot)	0.606	0.634
GPT-4o-mini (0-shot)	0.726	0.726
GPT-4o (0-shot)	0.756	0.755
o3 (0-shot)	0.753	0.763
DeepSeek (0-shot)	0.740	0.790
Llama-3 (4-shot)	0.491	0.694
Mistral (4-shot)	0.209	0.522
Mixtral (4-shot)	0.607	0.649
GPT-4o-mini (4-shot)	0.737	0.726
GPT-4o (4-shot)	0.742	0.725
o3 (4-shot)	0.742	0.760
DeepSeek (4-shot)	0.767	0.816
Human Upper Bound	0.834	0.892

Story	Word Sense	Human average	Model predictions
He spent years perfecting his craft. Every night, he rehearsed in his small, dimly lit room. Now, it was time to showcase his skills on the big stage. In front of the large audience, table set, he waited for his <u>cue</u> . He was ready to play snooker.	sports implement consisting of a tapering rod used to strike a cue ball in pool or billiards	2.8	[4, 2, 5, 4, 3]
It had been a long week for Emma. She felt overwhelmed by everything happening. On Saturday, she finally decided to do something about it. She took three <u>shots</u> that day.	a small drink of liquor	3.0	[4, 4, 3, 2, 2]
The girl packed her bag for a long adventure. She was excited to explore new places she had never been. Along the way, she faced unexpected challenges that left her feeling different. While on a backpacking trip, the girl ended up in a strange <u>state</u> . She hadn't been there before, but she liked it there and would go again.	the territory occupied by one of the constituent administrative districts of a nation	4.4	[5, 2, 2, 2, 3]

Table 4: Examples of human averages and LLM ratings for word sense plausibility in stories. All pictured model scores were taken from the zero-shot setting (order of scores: GPT 4o, 4o-mini, DeepSeek, Llama-3, Llama-3 + FT).

	Spearman	Acc. w/in SD
Random	0.000	0.454
Majority	N/A	0.558
Llama-3 (0-shot)	0.462	0.663
Llama-3+ FT (0-shot)	0.725	0.698
Mistral (0-shot)	0.382	0.568
Mistral + FT (0-shot)	0.692	0.726
Llama-3 (4-shot)	0.491	0.694
Llama-3 + FT (4-shot)	0.751	0.795
Mistral (4-shot)	0.209	0.522
Mistral + FT (4-shot)	0.684	0.733
Human Upper Bound	0.834	0.892

Table 5: Results for Llama-3 and Mistral out-of-the-box (LLama-3 / Mistral) and with additional fine-tuning (Llama 3 + FT / Mistral + FT).

3. Datasets

The training dataset is provided by the organizer [3]. The dataset is structured in the form of a list of json objects that contain the following fields:

- homonym: the word for which we are trying to compute the plausibility of word senses (the ambiguous word)
- judged_meaning: specifies which of the two candidate senses is being evaluated
- precontext: contains 3 sentences that establish the setting and narrative background
- sentence: an ambiguous fourth sentence that contains the homonym
- ending: provides additional narrative context that can influence the interpretation
- choices: a list of integer values containing the plausibility rating given by 5 annotators, each on a scale from 1 (very implausible) to 5 (very plausible) indicating how well the judged_meaning fits the story.
- average and stdev: summarizations of the choices field
- nonsensical: an array of boolean flags that have the value TRUE if the annotator judged the sentence to be nonsensical
- sample_id: integer value used to identify the training instance

The testing data contains instances of the same format, but the fields ‘choices’, ‘average’, ‘stdev’ and ‘nonsensical’ are empty.

Section	Avg. length (in words)
Precontext	31.5
Amb. Sentence	9.24
Ending	13.5
Entire Story	49.77

Table 2: Average length of story sections in our dataset.

We have 2 primary metrics for evaluation:

- Spearman Correlation: The predicted plausibility score should ideally correlate to the average of scores assigned by humans.
- Accuracy Within Standard Deviation: Some samples have a clearer consensus than others, so we calculate accuracy as the proportion of model predictions that are within standard deviation (at least 1) from the average judgment by annotators.

4. References

- [1] - <https://aclanthology.org/D09-1046.pdf>
(<https://scholar.google.com/scholar?q=Graded+Word+Sense+Assignment+Erk+McCarthy>)
- [2] - <https://aclanthology.org/D19-1355.pdf>
- [3] - <https://aclanthology.org/2025.starsem-1.12.pdf>